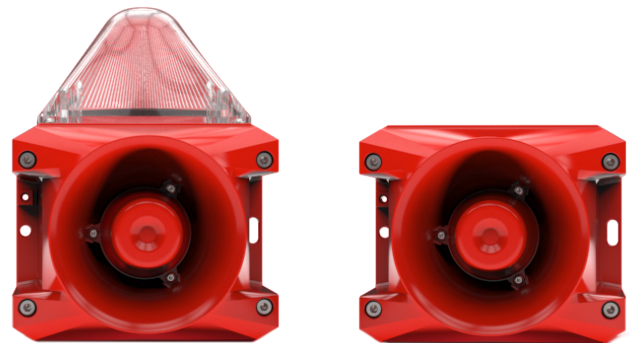


FUNCTIONAL SAFETY SIGNALING SOLUTIONS

When reliable signaling saves lives

What Is Functional Safety?

Functional Safety is the part of overall safety that depends on the correct operation of safety-related systems, for example, that a signaling device functions when a hazardous situation occurs. Defined in IEC 61508, it ensures that risks are reduced to an acceptable level through dependable control and warning systems.



Why Functional Safety Matters

A simple failure of a signaling device can result in the complete loss of the safety function. Only SIL/PL-certified signaling devices guarantee that even in the event of a fault the defined safety can be maintained and documented.

Risk Reduction

Failsafe signaling devices are designed to prevent accidents before they happen. Whether it's a start-up warning or a gas leak alert, these devices help to detect dangerous faults and ensure a safe operation.

Reliability

SIL (Safety Integrity Level) and PL (Performance Level) ratings aren't just technical jargon—they're proof of dependable operation. Certified devices behave consistently under stress, ensuring your safety systems work when they're needed most.

Compliance

Stay ahead of regulatory requirements with devices that meet global safety standards:

- **IEC 61508** – Functional safety of electrical/electronic/programmable systems
- **IEC 62061** – Safety of machinery using electrical control systems
- **ISO 13849** – Safety-related parts of control systems
- **EN 60204-1** – Electrical equipment of machines

Functional Safety Signaling Solutions

5 steps to select the right SIL/PL signaling device

- 1. Risk assessment and definition of required SIL/PL
- 2. Use verified data and manuals for safety loop calculation - Pfannenberg provides Safety Manuals, SISTEMA data and VDMA 66413 for easy integration
- 3. Definition of the danger zone based on risk assessment results
- 4. Determination of ambient noise level (for audible devices only) in danger zone
- 5. Selection and positioning of signaling devices to ensure perceptibility

Pfannenberg SIL/PL signaling devices

Engineered for Safety. Built for Performance.

- Safety according to IEC 61508 / ISO 13849
- Integrated monitoring functions for enhanced safety integrity
- Proven reliability - documented failure rates
- Safety documentation, SISTEMA library and VDMA 66413 Characteristic value library available for download
- Visible and audible even under demanding conditions

Practical Applications of SIL/PL Signaling Devices

	Start-up warning		Conveyor belt & loading area alerts
	Gas/leak / overpressure alarms		Muting indication during maintenance

Pfannenberg – Protecting people, machines, and the environment.

www.pfannenberg.com/functional-safety

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